

Xinyi Yang

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EDUCATION

Columbia University <i>M.S. in Artificial Intelligence; AI and Advanced Computing Concentration</i>	Sept.2026 – Dec.2027(Expected) New York City, USA
University of California, Santa Barbara <i>Exchange Student; Electrical Computer Engineering</i>	Sept.2025 – Jun.2026 Santa Barbara, USA
Huazhong University of Science and Technology <i>B.E. in Artificial Intelligence and Automation; Honors Class</i>	Sept.2022 – Jun.2026 Wuhan, China

RESEARCH & PROFESSIONAL EXPERIENCE

Intelligent Perception and Auto-Driving Lab, HUST <i>Research Assistant (Undergraduate Researcher)</i>	Jan. 2024 – Jan. 2025 Wuhan, China
- Developed a multimodal LLM agent by extending Monkey/Qwen-VL with external tool-use capabilities for perception, planning, and decision-making. - Built a GPT-4-based automated instruction-data generation pipeline to synthesize high-quality tool-calling conversations for instruction tuning. - Fine-tuned Qwen-VL with LoRA to enable external tool invocation, integrating Grounding DINO, SAM, and InstructPix2Pix into an end-to-end multimodal agent. - Contributed to the development and evaluation of a tool-augmented multimodal LLM that achieved 45.4 on MM-VET, outperforming the LLaVA-Plus baseline by 29.7%.	
CamAI+ Deep Learning and Neural Networks Program <i>Project Team Lead</i>	Jan. 2023 – Mar. 2023 Cambridge, Online
Developed a YOLOv5-based text-recognition pipeline; built a custom image-text dataset and evaluated performance under complex visual conditions.	
China Telecom Corporation Limited Wuhan Branch <i>AI Algorithm Intern</i>	Jun. 2025 – Sept. 2025 Jul. 2026 – Sept. 2026 Wuhan, China
Financial Document Intelligence (Summer 2025). Developed an intelligent document auditing system integrating sequence-to-sequence text generation with document understanding for automated financial document analysis. - Redesigned the generation pipeline, reducing repetitive outputs from 27% to 9%, and fine-tuned Faster R-CNN for document understanding, achieving 0.78 mAP. AI-based Examination Monitoring (Summer 2026). Developed a real-time examination monitoring system using YOLO11 for abnormal behavior detection in surveillance videos. - Optimized the detection pipeline with PyTorch to support efficient real-time inference in practical deployment scenarios.	

HONORS & AWARDS

Outstanding Completion, National Undergraduate Innovation and Entrepreneurship Training Program <i>National-level Undergraduate Research Project: Multimodal LLM Agent with Tool Use</i>	2025
Outstanding Student Scholarship, Huazhong University of Science and Technology	2023

SKILLS

Languages : Python, Matlab, C, R, SQL

Tools : Git, Conda, Pytorch, Jupyter Lab, Simulink, LaTeX, HuggingFace